MODULE 04

Data Structures & Algorithms

Turn problem-solving into a placement advantage

LEARN A SKILL • BUILD PROOF • CREATE YOUR NEXT OPPORTUNITY

Learn how strong developers think: choose the right structure, design an efficient solution and solve coding problems systematically.

WHO CAN JOIN?

Engineering students, freshers, placement seekers, career switchers with basic programming knowledge and coding-interview candidates.

WHY STUDENTS CHOOSE THIS PATH

- Beginner-friendly learning path — start with the fundamentals and progress step by step.
- Practical, project-focused training — learn by building instead of only watching lectures.
- Career-oriented skills — connect every topic to real applications and workplace use.
- Portfolio advantage — finish with projects you can discuss in interviews.
- A new direction is possible — your previous degree does not have to decide your entire career.

MODULE CONTENT

Time/space complexity	Big O/Ω/Θ
Arrays	Strings
Linked Lists	Stacks
Queues	Deques
Hashing	Binary Trees
BST	AVL
Heap	Priority Queue
Trie	Graphs
BFS/DFS	Shortest paths
Dijkstra	Bellman-Ford
MST	Kruskal/Prim
Topological Sort	Searching
Sorting	Recursion
Backtracking	Greedy
Divide & conquer	Dynamic Programming
Sliding Window	Two Pointers
Prefix Sum	Bit Manipulation
Coding Assessments	Mock Interviews

HANDS-ON PROJECTS

- Autocomplete System
- Route Finder
- Recommendation Engine
- Interview Problem Portfolio

CAREER DIRECTIONS

Software Developer • Product Company Candidate • Backend Developer • Coding Interview Candidate

YOUR JOURNEY

Learn → Practice → Build → Create a portfolio → Prepare for interviews → Apply your skills in real-world situations.

Note: Training improves skills and employability; employment depends on individual preparation, portfolio quality, interviews and available opportunities.